

RESEARCH INTERESTS

Robotics | Safety-Critical Control Systems | Autonomous Systems | Motion Planning & Navigation

EDUCATION

Degree	Institute	CGPA	Year of Graduation
Master of Science	University of Michigan, Ann Arbor	4.0/4.0 (current)	2027 (expected)
Bachelor of Technology	Indian Institute of Technology, Kharagpur	8.77/10	2025

TECHNICAL SKILLS

- Languages:** Python | MATLAB | C | Arduino | Visual Basic | LaTeX
- Software:** ROS | Gazebo | RViz | IPG CarMaker | dSPACE ControlDesk | MATLAB/Simulink | Fusion 360 | Pix4Dmapper

EXPERIENCE

- Graduate Research | Distributed Autonomous Systems and Control Lab | Prof. Dimitra Panagou** *September 2025 - Present*
 - Formulated a unified comparative framework to evaluate **Backup CBF**, **Model Predictive Shielding**, and **gatekeeper** filters
 - Implemented **SEAMLiS**, a modular execution-layer safety algorithm for decentralized **sensing-limited multi-robot exploration**
 - Developed an **Event-Triggered gatekeeper** to significantly reduce **computational costs** while preserving recursive feasibility
- Engineering Intern | ADAS Features Algorithm Development Team | Magna Electronics Inc.** *May 2026 - August 2026*
 - Optimized the **velocity controller** of the **Trained Parking Assist (TPA)** feature utilizing in-vehicle testing and log analysis
 - Improved the **curvature-based slowdown** of the **ACC velocity controller** and validated the behavior in **CarMaker** simulations
 - Integrated the **TPA** model with other low-speed features and validated back-to-back feature behavior in vehicle & simulation
- Undergraduate Research | Unmanned Aerial Vehicles Lab | Prof. Sikha Hota** *August 2024 - July 2025*
 - Implemented **Biased PNG**, **Two-Stage**, and **Time-to-go PNG** for 2D intercepts considering **acceleration and FOV constraints**
 - Adapted **multi-planar** guidance laws for **3D interception**, successfully eliminating lateral acceleration profile discontinuities
 - Designed a **2-planar 3-Staged 3D** Impact Angle Control Guidance (**IACG**) law that can achieve any terminal impact direction
- Research Intern | Control Law Group | RWR&DC | Hindustan Aeronautics Ltd** *May 2024 - July 2024*
 - Analyzed helicopter **AFCS** and integrated a **Pitt-Peters** dynamic inflow model to improve the transient response predictions
 - Validated the simulated transient responses against flight test data for **collective dump** inputs to identify future developments

PROJECTS

- Academic Project | Model Predictive Control | Prof. Ilya Kolmanovsky** *Jan 2026 - April 2026*
 - Engineered a **constrained MPC** using a **dual projected gradient QP** solver to achieve offset-free yaw rate tracking for vehicles
 - Integrated soft constraints via **slack variables** to ensure robust control performance against simulated plant-model mismatches
- Academic Project | Non Linear Control of an Inverted Pendulum | Prof. Inigo Incer** *Jan 2026 - April 2026*
 - Engineered a full-state feedback controller for a **nonlinear underactuated system** using dimensionless state transformations
 - Implemented a **Luenberger observer**-based dynamic compensator to estimate velocities and to verify closed-loop stability
- Unitree Go2 High-Speed Trajectory Controller | Dimensional OS (Open Source)** *Feb 2026*
 - Upgraded the **control architecture** from a 10Hz reactive model to a 100Hz setup with a **ROS** adapter for high-speed tracking
 - Designed an **adaptive Pure Pursuit** and **PID Cross-Track controller** alongside a novel **velocity profiler** to stop lateral drifts
- Academic Project | Online Learning for Control | Prof. Vasileios Tzoumas** *Sep 2025 - Dec 2025*
 - Implemented an **adaptive SSI-MPC** in **ROS2** utilizing Recursive Least Squares and predictors for unknown UAV dynamics
 - Integrated **Acados**-based solvers in a **Gazebo SITL** for high-speed multi-drone **trajectory tracking** against wind disturbances
- Academic Project | Aerospace Navigation and Guidance | Prof. Giusy Falcone** *Sep 2025 - Dec 2025*
 - Engineered a discrete-time **Kalman Filter** for accurate **state estimation** under a **Gaussian** process and measurement noise
 - Designed a computationally efficient **pursuit guidance law** for dynamic target interception in a high-speed engagement case
- Aerial 2D Mapping and Estimation of Forest Area Cover | Prof. Sandeep Saha** *Jan 2024 - Apr 2024*
 - Automated **semantic segmentation** and large-scale **environmental perception** via a **U-Net CNN** utilizing aerial drone imagery
 - Constructed maps from **telemetry** in **Pix4Dmapper**; applied **EDSR** to generate **Digital Surface Models** to estimate elevation
- Aerial Carbon Emission Monitoring System | Prof. Srinibas Karmakar** *Aug 2023 - Nov 2023*
 - Designed a drone-deployed embedded payload with **MQ7** and **MQ135** sensors, utilizing **nRF24L01** for real-time **RF** telemetry
 - Developed an **ascent time trajectory algorithm** and altitude **state estimation** via **BMP280** and dynamic atmospheric models

PUBLICATIONS

- T. Kim, **A. D. Menon**, A. Trivedi, and D. Panagou, "Backup-Based Safety Filters: A Comparative Review of Backup CBF, Model Predictive Shielding, and gatekeeper," *arXiv preprint arXiv:2604.02401*, 2026.
- T. Kim, R. H. Kumar, **A. D. Menon**, T.-H. Lin, and D. Panagou, "SEAMLiS: Visibility-Aware Safety for Perception-Limited Multi-Robot Exploration," *arXiv preprint arXiv:2607.09959*, 2026.
- N. K. Singh, S. Pal, **A. D. Menon**, and S. Hota, "Impact Angle Constrained Guidance Laws for Stationary Target Interception in 3D Space," in *AIAA SCITECH 2026 Forum*, 2026, p. 2348.

RELEVANT COURSEWORK

- Control & Robotics:** Model Predictive Control | Online Learning for Control | Motion Planning and Control | Non-Linear Systems and Control | Embedded Control Systems | Robotics | Systems Laboratory
- Dynamics & Guidance:** Aerospace Navigation and Guidance | Space Dynamics | Intermediate Dynamics | Flight Mechanics
- Flight Controls:** Flight Vehicle Controls | Flight Stability & Control | Automatic Control of Aircraft | Helicopter Engineering